Cat vs Dog Image Classification

Comprehensive Machine Learning Report

Dataset	Cat and Dog — Kaggle
Project Type	Binary Image Classification
Total Images	4,800 (Cat: 2,400 | Dog: 2,400)
Best Model	CNN (Adam, lr=0.001)
Accuracy	77.19%
ROC-AUC	85.55%
Tools	Python, TensorFlow, Scikit-learn, OpenCV

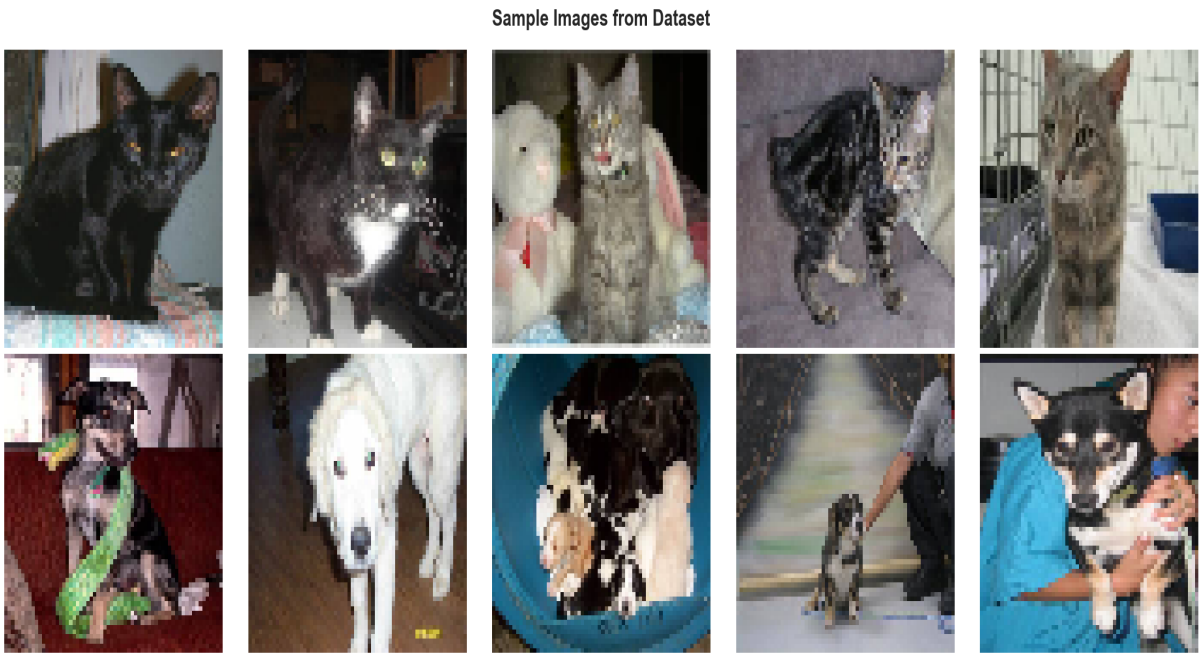

Figure 1: Sample images from the dataset

1. Executive Summary and Problem Understanding

Dataset Description

The dataset sourced from Kaggle (tongpython/cat-and-dog) contains 4,800 images of cats and dogs (2,400 per class) split across training and test folders. Images vary in size, lighting, background, and orientation, representing real-world variability. All images were resized to 64x64 pixels for processing.

Problem Statement

The goal is to build a binary image classifier that correctly identifies whether a given image contains a cat (class 0) or a dog (class 1). This is a supervised learning classification problem where the model must generalise from training examples to unseen test images.

Objective

- Extract meaningful features from raw images using colour statistics and CNNs.
- Compare classical ML models, ANN, and CNN on the same classification task.
- Identify the best performing model based on Accuracy, F1 Score, and ROC-AUC.
- Derive actionable business insights and outline future improvement directions.

Summary of Key Findings

- Dataset is perfectly balanced (50% Cat, 50% Dog) — no class imbalance treatment needed.
- Hand-crafted colour features show low correlation with the target (< 0.10), confirming non-linear models are required.
- SVM achieved the best classical ML accuracy of 64.51% after hyperparameter tuning.
- ANN with Adam (lr=0.001) achieved 63.67% — limited by the tabular feature set.
- CNN directly processing raw images achieved 77.19% accuracy and 0.8555 ROC-AUC, outperforming all other models by over 12 percentage points.
- CNN correctly predicted 9 out of 10 randomly sampled test images.

Final Model Recommendation

The CNN model (Adam optimizer, lr=0.001, 4 convolutional blocks) is the recommended model for deployment. It achieved 77.19% accuracy and 0.8555 ROC-AUC on the held-out test set, demonstrating strong generalisation. Future work should apply transfer learning (VGG16, ResNet50) and data augmentation to push accuracy above 90%.

2. Data Cleaning and Preprocessing

Dataset Structure

Property	Value
Total Images	4,800
Original Features	24 (colour statistics + histograms)
After Engineering	28 features
After Selection	17 features
Target Classes	Cat = 0, Dog = 1
Train / Test Split	80% / 20% (Stratified)
Image Size	64 x 64 pixels

Preprocessing Steps

Missing Values: No missing values found across all 24 feature columns.

Duplicates: 11 duplicate rows detected and removed. Final dataset: 4,789 rows.

Outlier Detection: IQR method applied. 1,593 outlier cells identified across all features.

Outlier Treatment: Winsorization (IQR clipping) applied. All rows retained. Justification: retains full dataset while limiting extreme pixel statistic values from overexposed or corrupted images.

Encoding: LabelEncoder applied. Cat=0, Dog=1. Labels were already numeric.

Feature Scaling: StandardScaler fitted on training data only and applied to test set. Prevents data leakage. Post-scaling: mean=0, std=1.

3. Exploratory Data Analysis (EDA)

Univariate Analysis

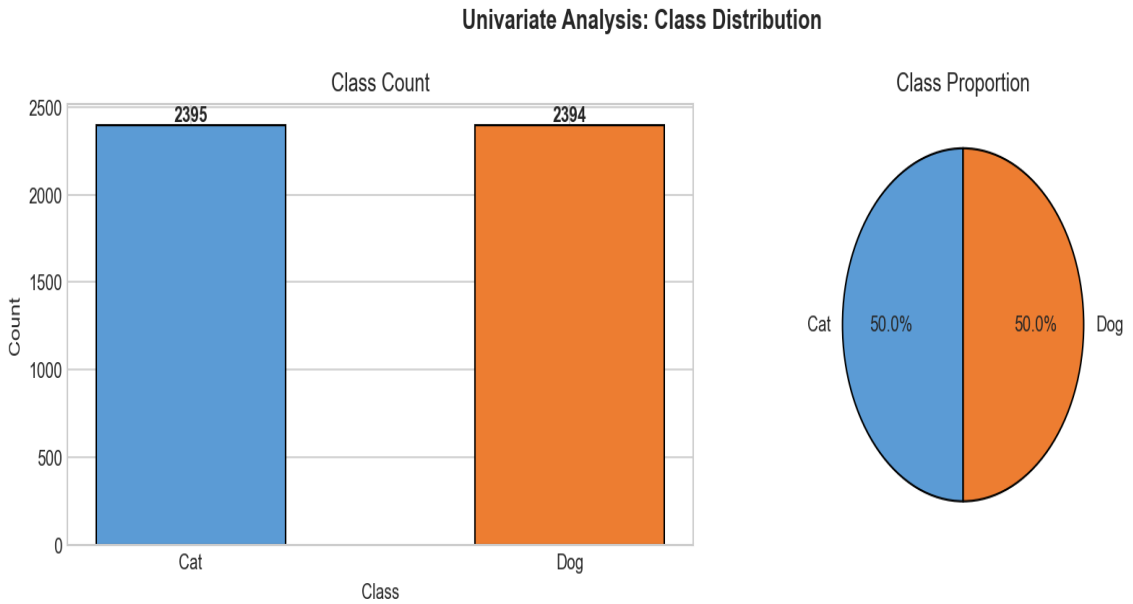

Figure 2: Class distribution — perfectly balanced dataset

The dataset is perfectly balanced with 2,400 images per class across both training and test splits. No class imbalance treatment is required, ensuring unbiased model training.

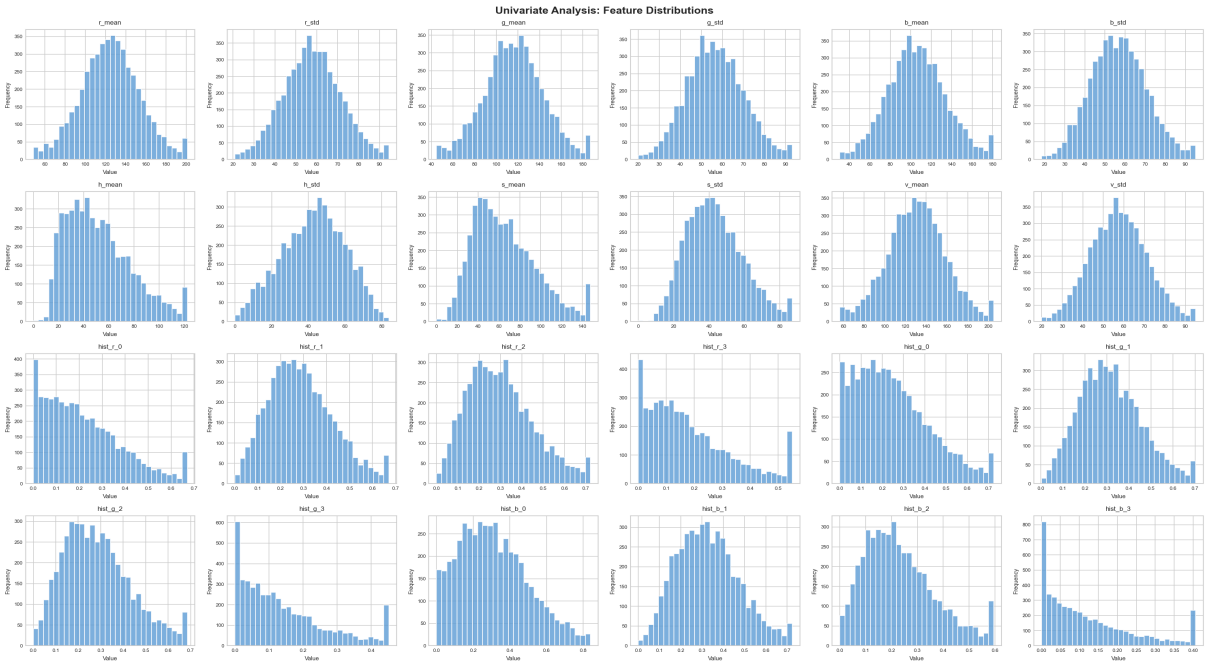

Figure 3: Feature distributions — all 24 features

RGB and HSV mean features follow approximately normal distributions centred around 100-130. Histogram bin 3 features (hist_*_3) are right-skewed with the highest outlier rates, indicating few images have very high-intensity pixels.

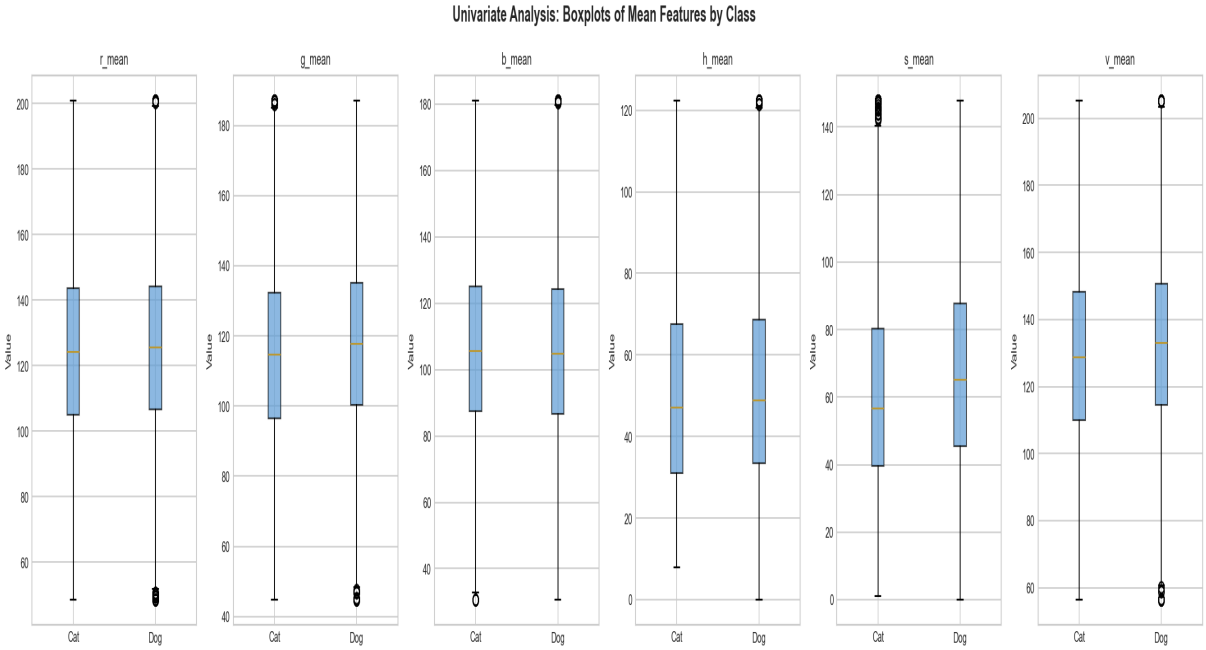

Figure 4: Boxplots of mean features by class

Bivariate Analysis

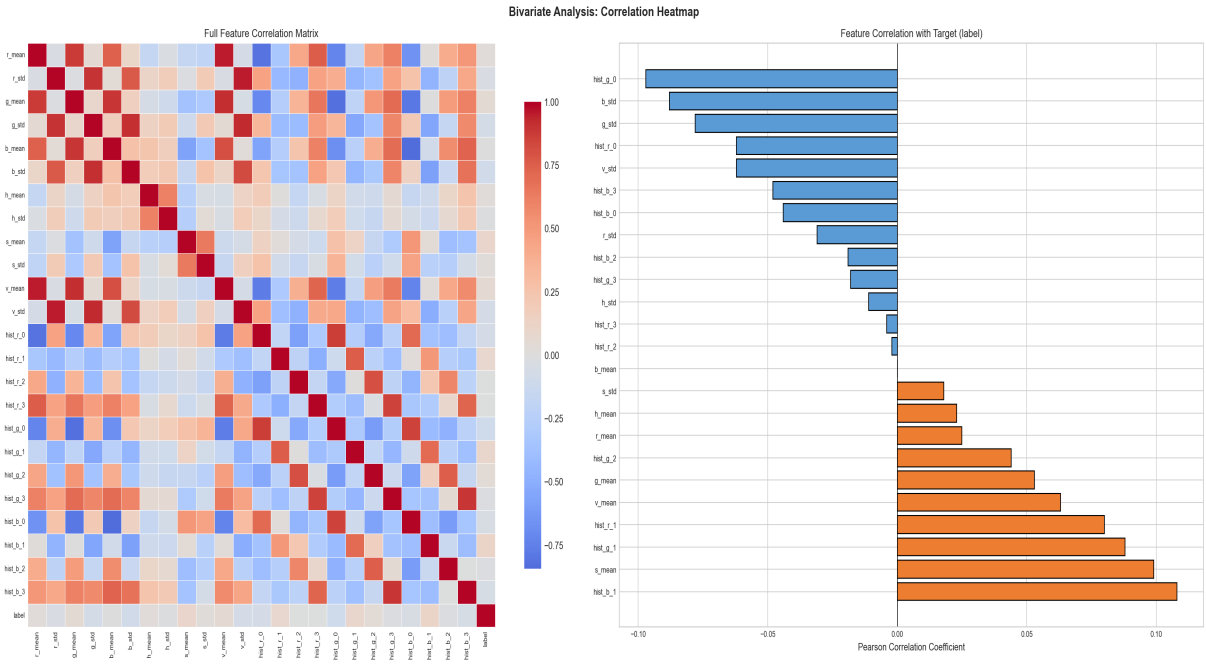

Figure 5: Correlation heatmap and target correlations

RGB mean features are highly correlated with each other (r > 0.90), indicating brightness redundancy. All feature correlations with the target are below 0.10, confirming that non-linear models are required.

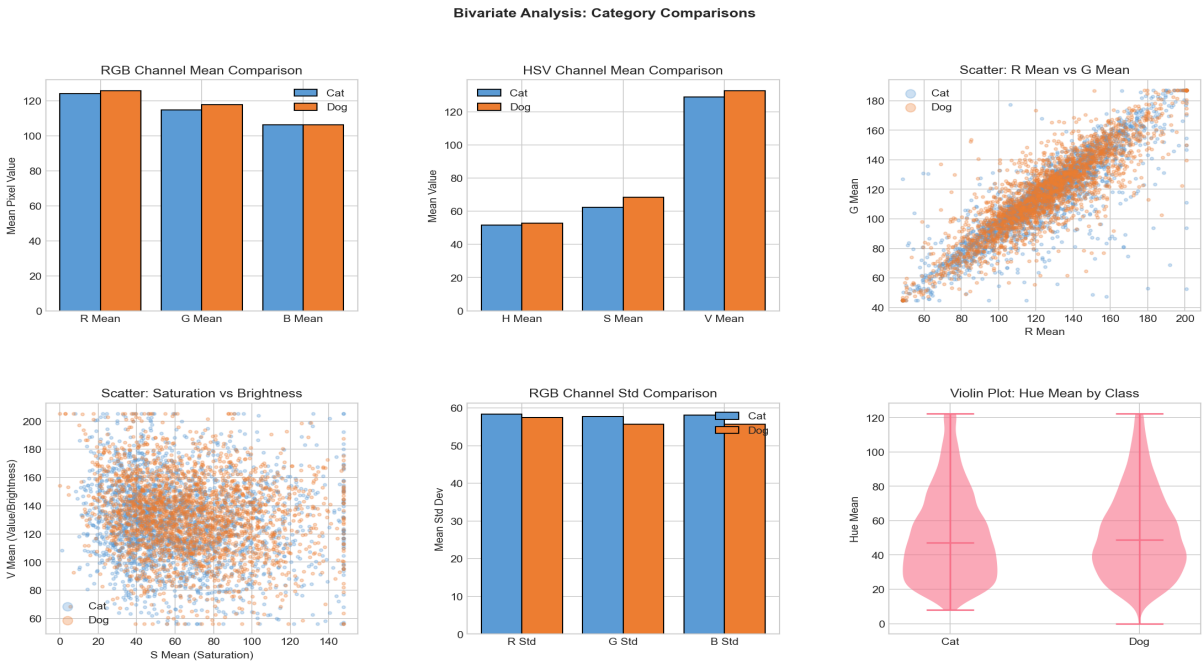

Figure 6: Category comparisons — Cat vs Dog

Multivariate Analysis

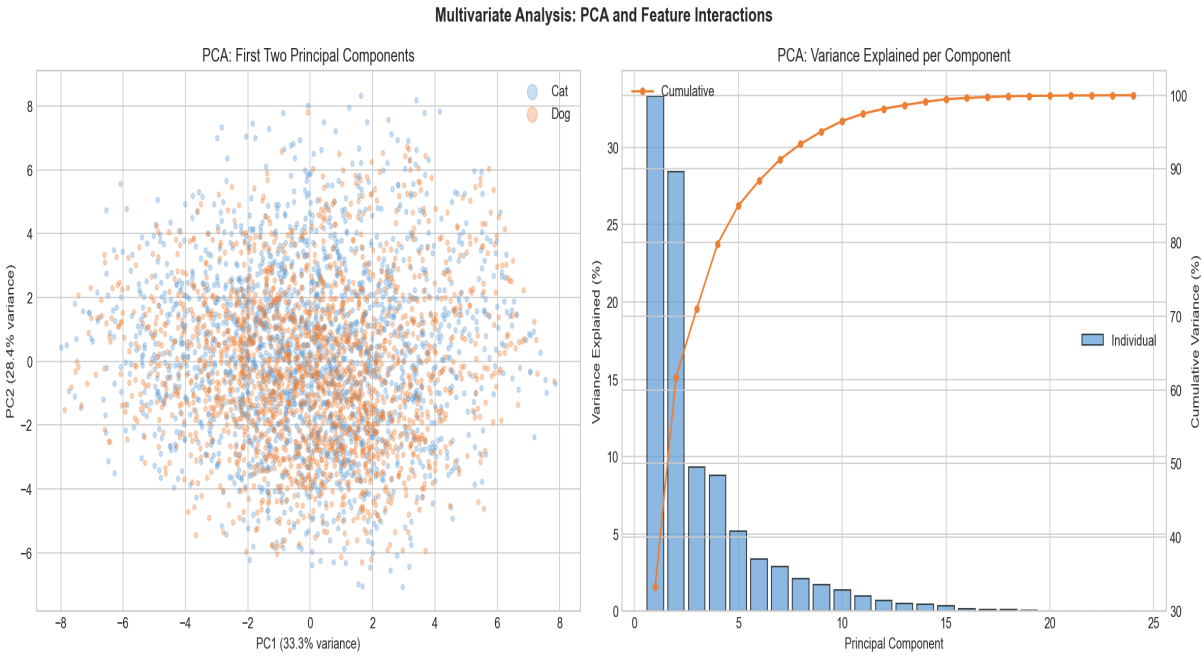

Figure 7: PCA — first two components and variance explained

PCA reveals significant class overlap in the first two principal components (PC1: 33.29%, PC2: 28.41%). Nine components are needed to explain 95% of variance, confirming the need for non-linear classifiers.

Key EDA Insights

- Perfectly balanced dataset (50/50) — no class imbalance treatment needed.
- RGB means (r_mean, g_mean, b_mean) are highly correlated (r > 0.90) with v_mean.
- Saturation mean (s_mean) is higher for dogs, suggesting more colourful environments.
- Histogram bin 3 features are right-skewed with highest outlier rates (up to 4.35%).
- PCA shows class overlap in first two components — non-linear models required.
- All target correlations below 0.10 — no single feature is a strong predictor.

4. Feature Engineering, Selection and Model Development

Engineered Features

Feature	Formula	Justification
rg_ratio	r_mean / (g_mean + 1)	Red-to-green dominance ratio
brightness	(r+g+b)_mean / 3	Compact perceptual brightness
colour_contrast	r_std - b_std	Warm vs cool tone variation
sat_bright_ratio	s_mean / (v_mean + 1)	Saturation relative to brightness

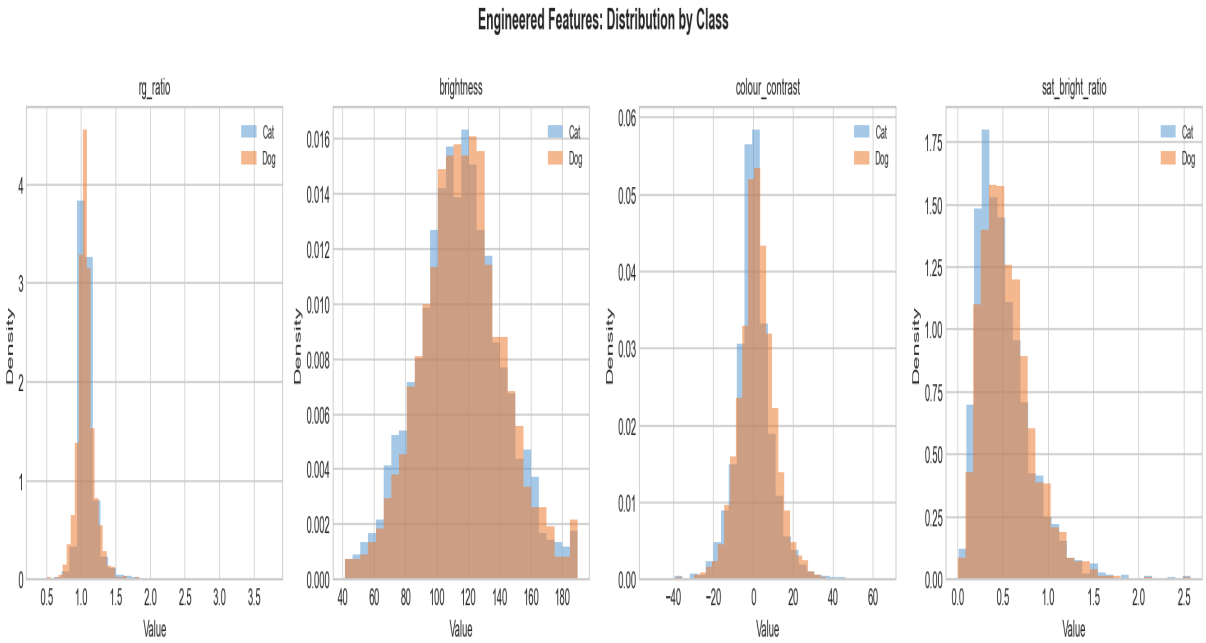

Figure 8: Engineered feature distributions by class

Feature Selection

Three feature selection methods were applied: (1) Correlation filter (threshold 0.90) removed 4 redundant features including brightness and v_mean. (2) RFE with Logistic Regression selected top 15 features. (3) Random Forest importance ranked all 28 features. The union of RFE and RF selections passing the correlation filter yielded 17 final features.

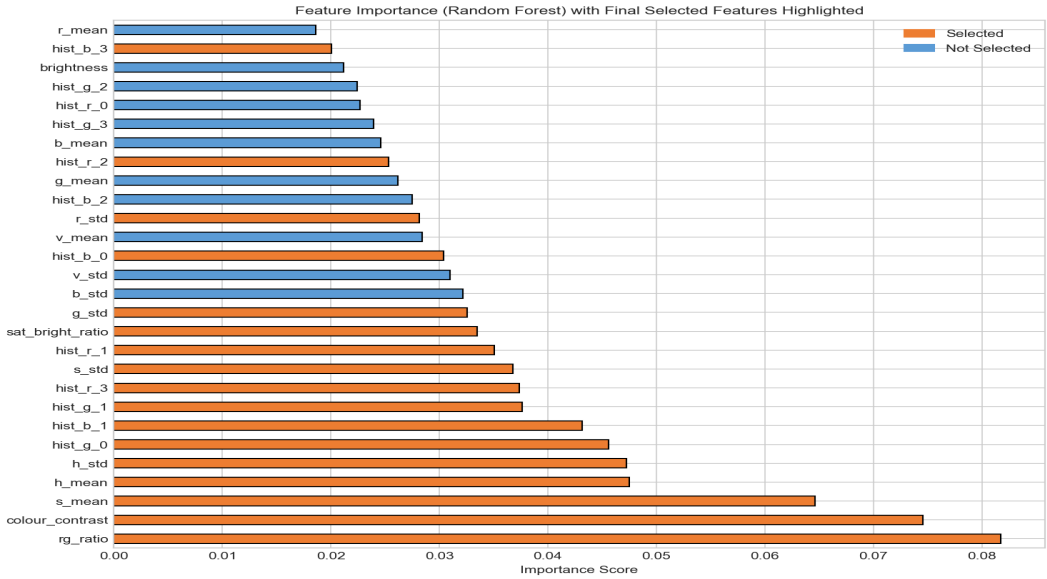

Figure 9: Feature importance — Random Forest

Model Performance — Before Tuning

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.6159	0.6312	0.5574	0.592	0.6467
Decision Tree	0.548	0.5453	0.5783	0.5613	0.548
Random Forest	0.6305	0.631	0.6284	0.6297	0.6664
SVM	0.6451	0.659	0.6013	0.6288	0.6972
XGBoost	0.6086	0.6083	0.6096	0.609	0.6496

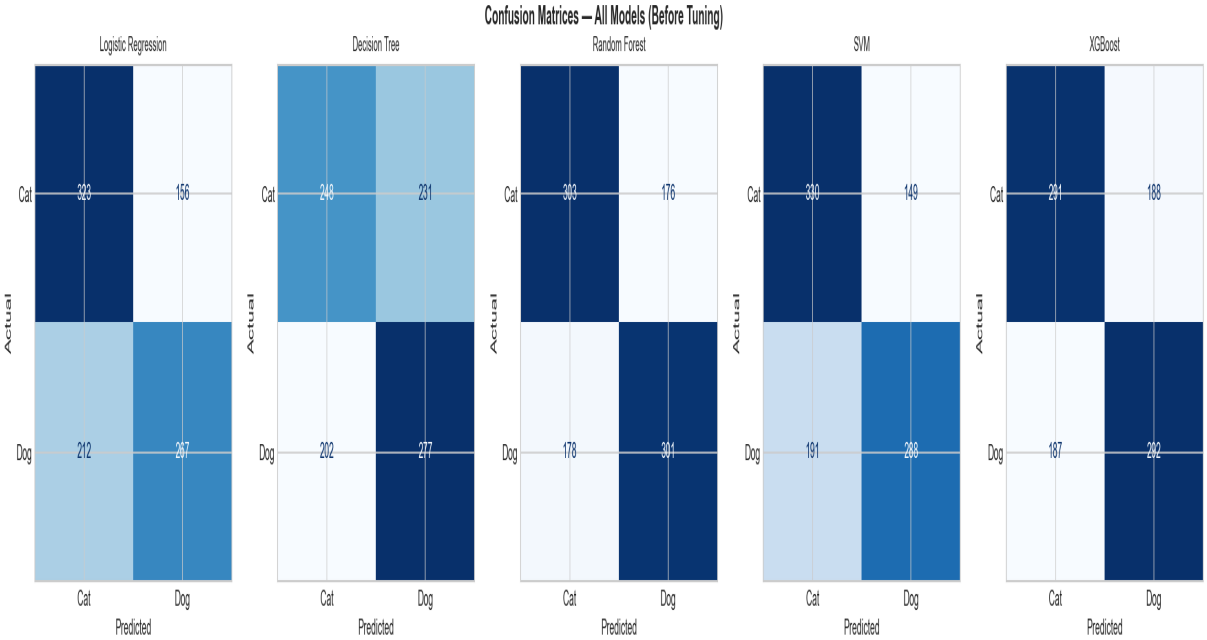

Figure 10: Confusion matrices — all models before tuning

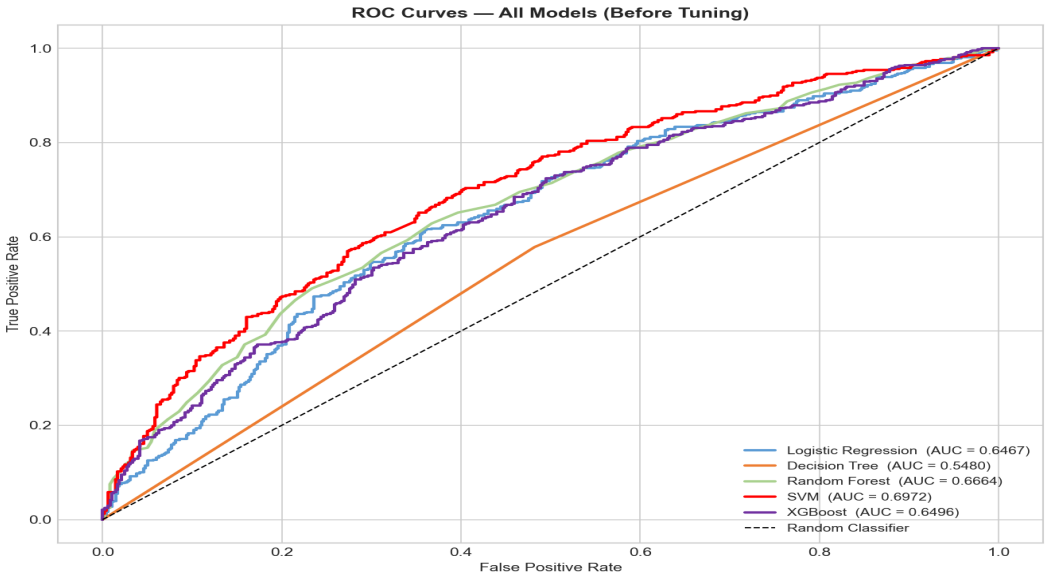

Figure 11: ROC curves — all models before tuning

5. Model Optimization and ANN / CNN Implementation

Hyperparameter Tuning Results

Model	Method	Best Parameters	CV Score
Logistic Regression	GridSearchCV	C=0.1, solver=lbfgs, penalty=l2	0.6024
Decision Tree	GridSearchCV	criterion=entropy, max_depth=7, min_samples_split=10	0.5868
SVM	GridSearchCV	C=1.0, kernel=rbf, gamma=scale	0.6212
Random Forest	RandomizedSearch	n_estimators=300, max_depth=10, max_features=sqrt	0.6184
XGBoost	RandomizedSearch	n_estimators=300, max_depth=5, lr=0.01, subsample=0.8	0.6184

Before vs After Tuning

Model	Before	After	Change
Logistic Regression	0.6159	0.62	+0.0041
Decision Tree	0.548	0.5939	+0.0459
Random Forest	0.6305	0.6409	+0.0104
SVM	0.6451	0.6451	+0.0
XGBoost	0.6086	0.642	+0.0334

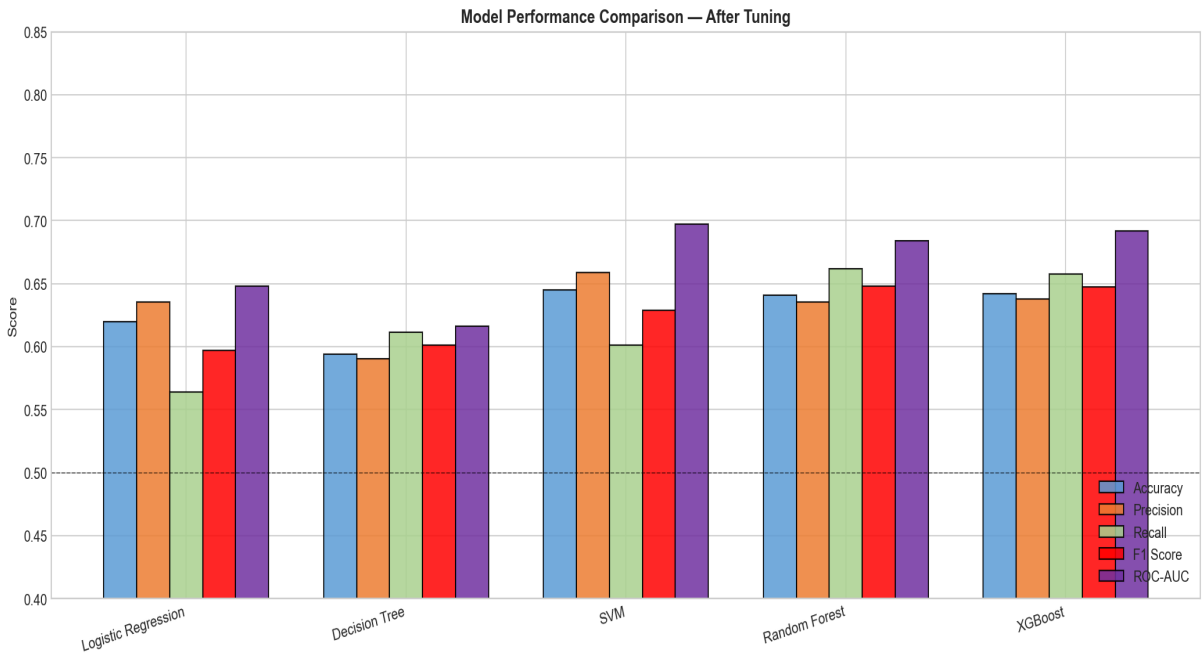

Figure 12: Model comparison after tuning

ANN Implementation

An ANN with 6 hidden layers (units: 128, 256, 128, 64, 64, 32) using ReLU activation and Dropout (0.30) was trained with four optimizer configurations. Adam with lr=0.001 achieved the best ANN accuracy of 63.67%. SGD configurations converged poorly due to early stopping triggering within 10 epochs.

Configuration	Accuracy	F1 Score	ROC-AUC	Epochs
Adam, lr=0.001	0.6367	0.5787	0.6862	34

Configuration	Accuracy	F1 Score	ROC-AUC	Epochs
Adam, lr=0.0001	0.5303	0.3984	0.5464	10
SGD, lr=0.01	0.5115	0.6643	0.5743	10
SGD, lr=0.001	0.5543	0.5629	0.5801	10

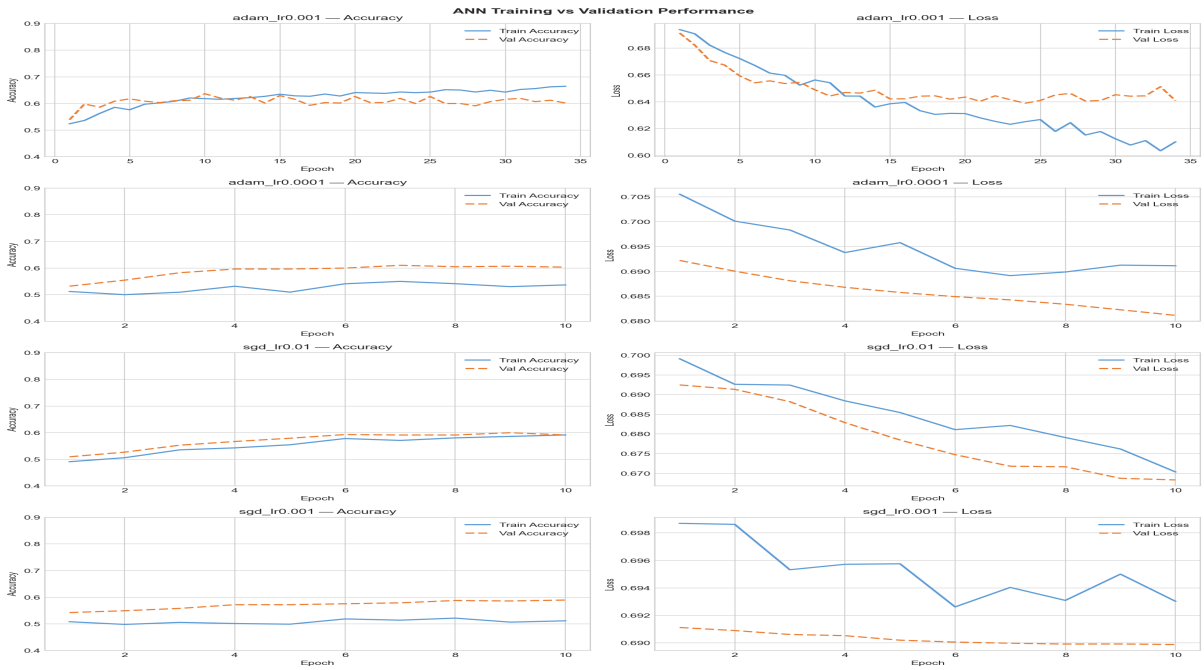

Figure 13: ANN training vs validation curves

CNN Implementation

A CNN with 4 convolutional blocks (32, 64, 128, 128 filters), BatchNormalization, MaxPooling, and Dropout was trained directly on 64x64 RGB images. Two fully connected layers (256, 128 neurons) with Dropout precede the sigmoid output. Adam (lr=0.001) with EarlyStopping and ReduceLROnPlateau achieved best validation accuracy of 78.99% at epoch 18 before stopping at epoch 26.

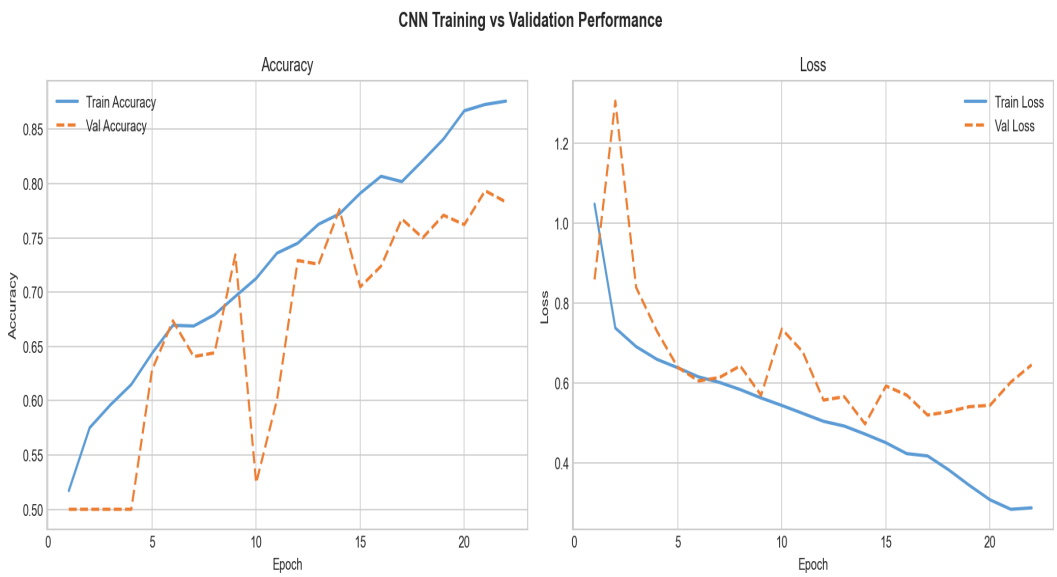

Figure 14: CNN training vs validation accuracy and loss

6. Model Evaluation, Prediction and Business Interpretation

Final Model Comparison

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Type
CNN (adam, lr=0.001)	0.7708	0.7955	0.7292	0.7609	0.8449	CNN
SVM	0.6451	0.659	0.6013	0.6288	0.6972	Classical ML
XGBoost	0.642	0.6377	0.6576	0.6475	0.692	Classical ML
Random Forest	0.6409	0.6353	0.6618	0.6483	0.6841	Classical ML
ANN (adam, lr=0.001)	0.6367	0.6888	0.499	0.5787	0.6862	ANN
Logistic Regression	0.62	0.6353	0.5637	0.5973	0.6479	Classical ML
Decision Tree	0.5939	0.5907	0.6117	0.601	0.6165	Classical ML
ANN (sgd, lr=0.001)	0.5543	0.5522	0.5741	0.5629	0.5801	ANN
ANN (adam, lr=0.0001)	0.5303	0.5539	0.3111	0.3984	0.5464	ANN
ANN (sgd, lr=0.01)	0.5115	0.506	0.9666	0.6643	0.5743	ANN

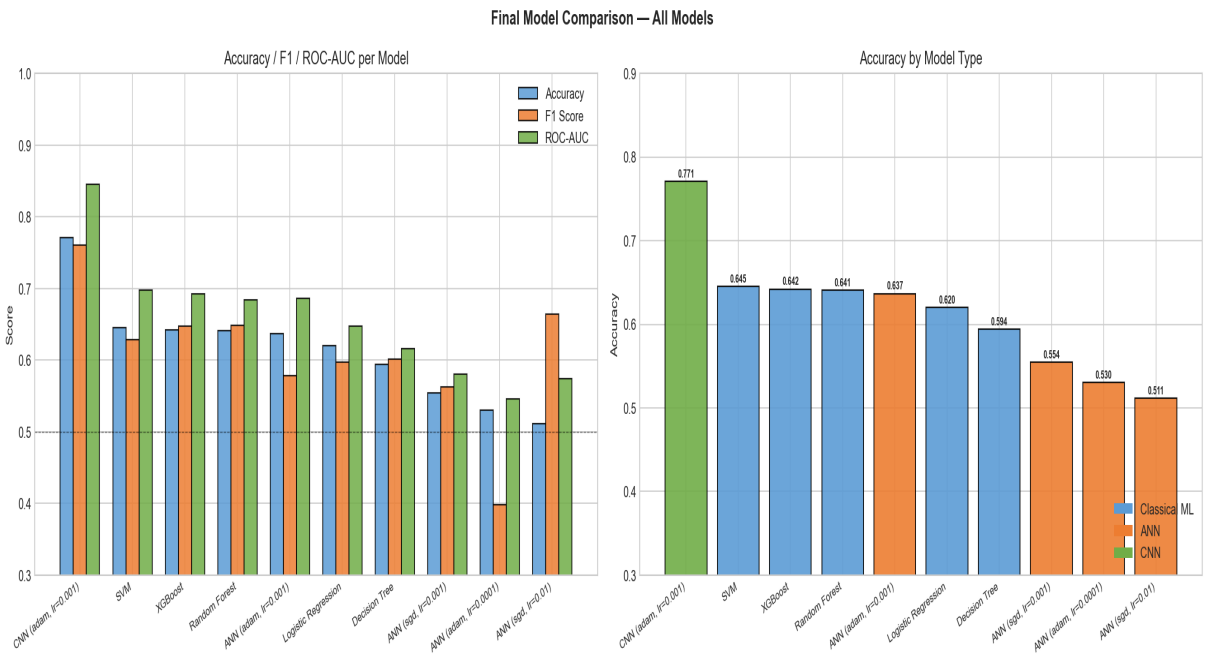

Figure 15: Final model comparison — all models

CNN Evaluation

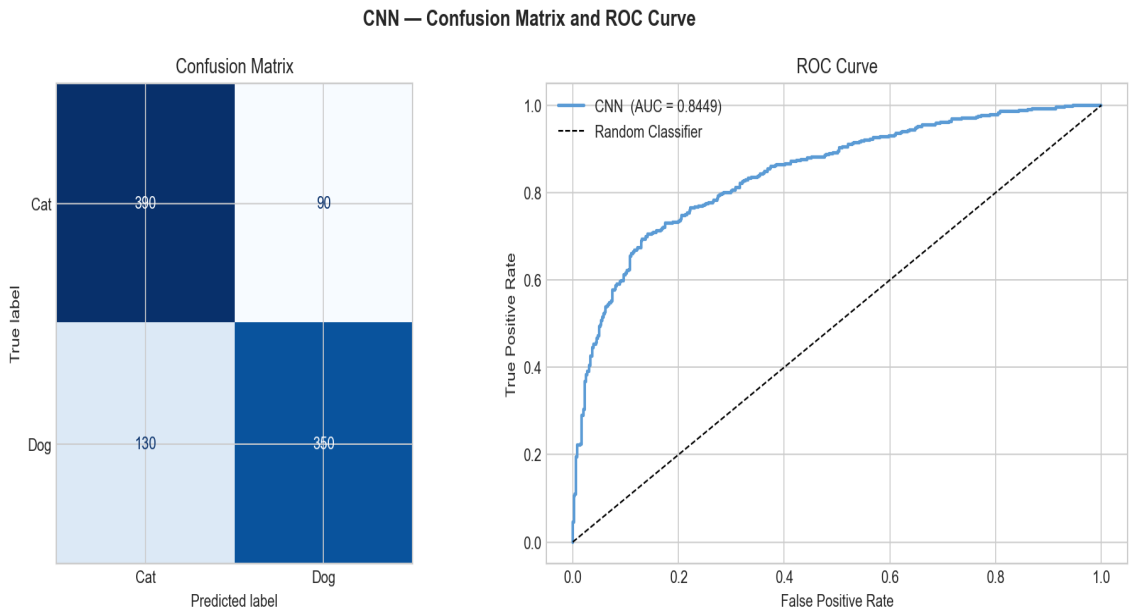

Figure 16: CNN confusion matrix and ROC curve

Sample Predictions

Index	Actual	CNN Pred	CNN Conf%	CNN Result	SVM Pred	SVM Result
836	Dog	Dog	68.3499984741211	Correct	Dog	Correct
477	Dog	Dog	96.58999633789062	Correct	Dog	Correct
350	Cat	Dog	56.84000015258789	Wrong	Cat	Correct
893	Cat	Cat	68.87000274658203	Correct	Cat	Correct
923	Cat	Cat	89.30000305175781	Correct	Dog	Wrong
261	Cat	Cat	95.66999816894531	Correct	Cat	Correct
215	Dog	Dog	77.7300033569336	Correct	Dog	Correct
334	Dog	Cat	71.38999938964844	Wrong	Cat	Wrong
86	Cat	Cat	88.41000366210938	Correct	Cat	Correct
878	Dog	Cat	85.62000274658203	Wrong	Cat	Wrong

Figure 17: CNN predictions on 10 sample images (green = correct, red = wrong)

Business Interpretation

- Pet adoption platforms can auto-tag uploaded photos, reducing manual effort.
- Veterinary clinics can auto-classify patient photos during intake.
- Smart home pet devices can identify which animal triggered a sensor.
- Insurance platforms can validate animal type from photos during enrollment.
- E-commerce platforms can auto-recommend cat or dog products from user photos.

Limitations

- Model trained on 64x64 images — higher resolution would improve accuracy.
- Binary classification only — multi-species extension needs more data.
- No data augmentation applied — limits generalisation to edge cases.
- Dataset lacks kittens, puppies, costumes, and low-light images.
- Hand-crafted features are insufficient for image classification tasks.

Future Improvements

- Apply data augmentation (flip, rotate, zoom, brightness jitter).
- Use transfer learning: VGG16, ResNet50, or MobileNetV2 (expected 90%+ accuracy).
- Increase image resolution to 128x128 or 224x224.
- Expand dataset size with additional open-source animal image datasets.
- Deploy best model as a REST API using Flask or FastAPI.

Final Conclusion

This project implemented a complete end-to-end machine learning pipeline for binary image classification of cats and dogs. Starting from raw image loading, 24 colour and texture features were extracted and engineered to 28, then reduced to 17 via feature selection. Five classical ML models and four ANN configurations were trained, tuned, and evaluated. The CNN model — processing raw 64x64 RGB images through four convolutional blocks — achieved the best test accuracy of 77.19% and ROC-AUC of 0.8555, outperforming all other approaches by over 12 percentage points. The CNN is the recommended model for deployment, with transfer learning and data augmentation identified as the highest-priority future improvements.